

DATA STRUCTURE

LAB PROGRAM

1. Write C program to perform Matrix Multiplication (2D Array)

```
#include <stdio.h>

int main() {
    int a[10][10], b[10][10], c[10][10];
    int r, col, i, j, k;

    printf("Enter rows and columns: ");
    scanf("%d %d", &r, &col);

    printf("Enter first matrix:\n");
    for(i=0;i<r;i++)
        for(j=0;j<col;j++)
            scanf("%d",&a[i][j]);

    printf("Enter second matrix:\n");
    for(i=0;i<r;i++)
        for(j=0;j<col;j++)
            scanf("%d",&b[i][j]);

    for(i=0;i<r;i++) {
        for(j=0;j<col;j++) {
            c[i][j]=0;
            for(k=0;k<col;k++)
```

```

        c[i][j] += a[i][k] * b[k][j];
    }
}

printf("Result Matrix:\n");
for(i=0;i<r;i++) {
    for(j=0;j<col;j++)
        printf("%d ", c[i][j]);
    printf("\n");
}
return 0;
}

```

Enter rows and columns: 3 3

Enter first matrix:

2 3 4

2 3 8

6 0 7

Enter second matrix:

3 5 8

0 0 1

7 3 1

Result Matrix:

34 22 23

62 34 27

67 51 55

=== Code Execution Successful ===

1. Write C program to perform Matrix Multiplication (3D Array)

```
#include <stdio.h>
```

```

int main() {
    int a[2][2][2], c[2][2];
    int i, j, k;

    printf("Enter first matrix:\n");
    for(i=0;i<2;i++)
        for(j=0;j<2;j++)
            scanf("%d",&a[0][i][j]);

    printf("Enter second matrix:\n");
    for(i=0;i<2;i++)
        for(j=0;j<2;j++)
            scanf("%d",&a[1][i][j]);

    for(i=0;i<2;i++) {
        for(j=0;j<2;j++) {
            c[i][j]=0;
            for(k=0;k<2;k++)
                c[i][j] += a[0][i][k] * a[1][k][j];
        }
    }

    printf("Result Matrix:\n");
    for(i=0;i<2;i++) {
        for(j=0;j<2;j++)
            printf("%d ", c[i][j]);
        printf("\n");
    }
}

```

```
    return 0;  
}
```

Enter first matrix:

1 2

2 3

Enter second matrix:

1 2

2 7

Result Matrix:

5 16

8 25

=== Code Execution Successful ===